

Title & Core Problem Statement

Title	Cognitive Bias-Aware Negotiation Agents with Multimodal Debiasing
Problem Statement	The challenge is to design Reinforcement Learning (RL)-based negotiation agents that can detect and counteract anchoring biases in human negotiation partners.
Objective	The agent must strategically modify its speech patterns and response timing to either leverage or mitigate detected human bias for optimal negotiation outcomes.
Key Biases Addressed	The agent is designed to detect and manage biases such as anchoring bias
Novel Mechanism	Utilizing strategic speech generation parameters (tone, pace, word choice) as the action space for debiasing.

The Research Gap and Novelty

Category	Description and Context
Traditional AI Focus (The Gap)	Current negotiation AI research primarily focuses on strategy optimization (e.g., maximizing profit or getting the best deal). They often fail to model human cognitive biases as a first-class element of the negotiation strategy.
Novel Approach (Shift in Focus)	This research shifts the paradigm from purely rational strategy optimization to bias-aware, psychology-driven negotiation. The agent learns not just what to say, but how to influence the human partner.
Multimodal Innovation	Very few, if any, existing models incorporate multimodal debiasing (manipulating tone, prosody, and timing) as an explicit component of the negotiation strategy.

Multi-Bias Detection Framework

Detection Component	Mechanism and Features Analyzed
ASR Analysis	Automatic Speech Recognition (ASR) is used to analyze the human partner’s speech.
Speech Prosody	Analyzing features like tone, pitch, intonation, and stress patterns to identify confidence levels or hesitation.
Pause Patterns	Detecting and analyzing long silences and hesitations which can signal doubt, lack of confidence, or reconsideration of an anchor.
Linguistic Markers	Analyzing word choice and linguistic markers that indicate specific biases, such as firm words used in overconfidence ("I am 100% sure"), or arguments supporting one's own view ("All my friends also got it for \$100").
Example Detection	A long pause combined with an uncertain tone can signal a lack of confidence, often associated with anchoring bias following an initial price set by the seller.

Strategic Bias Exploitation vs. Mitigation

RL Decision Point	Description and Examples
Reinforcement Learning (RL)	The RL agent is trained to decide between two core actions: exploiting a bias or debiasing/mitigating it.
Reward Signals	The agent learns this policy by receiving reward signals based on various negotiation outcomes, such as profit maximization, agreement reached, human partner satisfaction, or long-term trust building.
Exploitation Strategy	If a human shows anchoring bias at a low price (e.g., anchored at \$100), the agent may exploit it by offering a price slightly higher (\$120) but still much lower than market value (\$150), maximizing profit under the existing low anchor.
Mitigation (Debiasing) Strategy	If a bias harms fairness or trust, the agent may debias it. This involves using objective criteria (e.g., reminding the human of market value or providing evidence) to counteract the biased belief.

Multimodal Action Space (Debiasing Tools)

Action Space Parameter	Strategic Use and Impact
Tone	The agent can adjust its tone (e.g., calm vs. assertive). If the human is overconfident, the agent might adopt a calmer tone.
Pace (Speed)	The pace of speech can be adjusted; speaking fast may push urgency or indicate confidence, while speaking slow might calm the partner or emphasize critical evidence.
Word Choice	Agent chooses between logical, evidence-based language (used to mitigate bias) versus emotional appeals or hedging words ("perhaps," "let's consider") used to introduce doubt.
Response Timing	The agent controls the time between turns, using an immediate reply versus a deliberate pause for effect. A pause, for example, might be used to make a human's previously stated anchor feel less certain.
Integrated Example	If the human exhibits overconfidence, the agent responds with a slower, calm tone while using evidence-based arguments to subtly introduce doubt.

Example Negotiation Flow (Bias-Driven)

Step	Interaction and Agent Analysis
Human Input	Buyer: "Hmm... okay, maybe ₹10,000?" (accompanied by a long pause and hesitant tone).
AI Detection	Agent detects Anchoring Bias (buyer is still close to seller's initial hypothetical price) and a Weakening Stance (hesitation/uncertainty).
RL Action	The RL policy dictates Exploitation of the uncertainty. The agent adjusts its multimodal output:
AI Response	Agent speaks with a confident, strong tone, at a slightly faster pace, and uses firm language: "Actually, ₹14,500 is already a discount from the market average of ₹15,000.".
Result	The agent actively used multimodal features (tone, pace) to maintain the high anchor and push the human toward a profitable agreement.